

Item	Section	Component	Ref	Value	Qty	Notes
1	Logic Core	ESP32-C3-DevKitM-1 (dev board)			1	Brain, Wi-Fi/NTP
2	Logic Core	DS3231 RTC module (w/ AT24C32 + CR2032)			1	Check onboard pull-up (likely already present)
3	Logic Core	AHT25 module			1	Check onboard pull-up (photo suggests it does not have one)
4	Logic Core	MAX7219 (IC or module)			1	Display controller, SPI
5	Display Power St	74HC540			1	Octal inverter (DIG logic)
6	Display Power St	UDN2981 (or TD62783)			1	12V source driver (segments)
7	Display Power St	ULN2803			1	Sink driver (common cathode/digit)
8	Displays	WCN1-00D0SR-C11S (4", Super Red, CC)			6	VF seg ~9.5V, VF DP ~1.9V
9	Displays	WCN1-00B2SR-C11S (2.3", Red, CC)			2	VF seg ~8.0V, VF DP ~4.0V
10	Resistors	Resistor 1/4W	R1	33R	7	Main A-G current limiter (4" display bus)
11	Resistors	Resistor 1/4W	R2	47R	7	Series A-G trim, only on the 2x 2.3" displays
12	Resistors	Resistor 1/4W	R3	390R	1	DP limiter (single bus, all digits)
13	Resistors	Resistor (pull-up SDA)	R4	4.7k	1	Only if multimeter test confirms absence on AHT25/DS3231
14	Resistors	Resistor (pull-up SCL)	R5	4.7k	1	Same as above